

Gen-AI training pre-requisites and pre-read

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Install and setup Python

1. Install Python

<https://www.python.org/downloads/release/python-3127/>

Use Python 3.12.6 or 3.12.7 only as it is tested with the libraries we will work on
DO NOT INSTALL LATEST PYTHON VERSION

When installing make sure the “Add python to system path” checkbox is selected

2. Test python

Open a command prompt and give the following command:

```
python --version
```

It should show the python version, namely, 3.12.7

If it errors finding the executable, add python.exe location to System environment variable: PATH

You can find the location of python.exe by opening the windows menu, right click on Python app (open file location), copy path

Install and setup VS Code

1. Download and install [VS Code](https://code.visualstudio.com/)

<https://code.visualstudio.com/>

2. Install python extensions in VS Code

1. Open VS Code
2. Install the Python extension:
 - Click the Extensions icon in the sidebar (or press Ctrl+Shift+X)
 - Search for "Python"
 - Install the Microsoft Python extension
 - Other VS Code extensions that are needed only if you want to run the Python notebooks in VS Code (you can always run them in colab)
Jupyter, Jupyter Notebook renderers, ipykernel

Understanding and creating a Virtual Environment for Gen AI projects

Understanding Project Root folder

Project root folder (such as **genaitraining**) or any folder in which you are putting your code, provides the base folder to work against.

When opening VS Code, this is the folder you will open (not any other subfolder)

This is important as your project virtual environment, private configuration (.env) will all reside in your project root folder.

Hence, whenever starting VS Code, **always open the project root folder and not any other folder**

Create a virtual environment for your project

What is Virtual Environment?

Virtual environment facilitates creation of a separate folder where all your python dependencies are saved. This ensures that you do not have to reinstall any library everytime and also that the right versions of your libraries are available to your application code (no conflicts with any other project or setup on your machine)

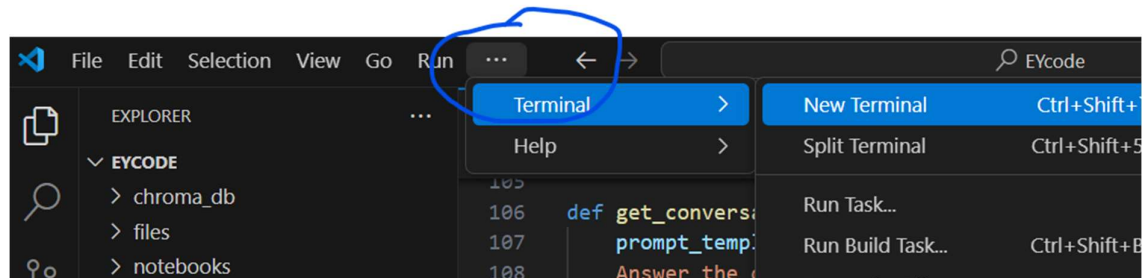
Following video explains the concept and benefits of virtual environments:

<https://youtu.be/KxvKCSwlUv8?si=6HQHCEJ2wkXnJ0QX>

Follow along the video below or steps outlined below to create a virtual environment

<https://youtu.be/GZbeL5AcTgw?si=cvCzwnbiXPZEWFMA>

1. Create a new folder for your project, called **genaitraining (or code)** in your home directory or in Desktop
2. Open the folder in VS Code (File > Open Folder)
3. Start a terminal



4. In the terminal window, give the following commands to create a new virtual environment:

```
python -m venv venv
```

```
venv\Scripts\activate
```

You will see a popup asking you to use this virtual environment for your project. Click on "Yes"

Install packages needed for Gen AI development

Python package manager **pip** allows you to install all needed packages from a file commonly named as **requirements.txt**

We will create requirements.txt and put all the packages we need for Gen AI development in it. We will then install those packages in our virtual environment.

1. In VS Code, with the project root folder (genaitraining or code) selected, click on the +File icon to create a new file, name it “requirements.txt”
2. Put following content in requirements.txt and save the file (ctrl+s):

```
langchain-openai  
langchain-community  
langchain-text-splitters  
chromadb  
beautifulsoup4  
pypdf  
wikipedia  
plotly  
tqdm  
datasets  
python-dotenv  
openai  
streamlit  
dotted-dict  
PyPDF2  
matplotlib  
seaborn  
torch  
transformers  
tensorflow  
tf-keras  
jupyter  
ipykernel
```

3. Install packages from requirements.txt:

Make sure your virtual environment is active (you should see the venv name in the command prompt)

```
14 "AZURE_OPENAI_ENDPOINT": os.getenv("AZURE_OPENAI_ENDPOINT"),
15 "AZURE_OPENAI_MODEL_DEPLOYMENT_NAME": os.getenv("AZURE_OPENAI_MODEL_DEPLO
16 "AZURE_OPENAI_MODEL_NAME": os.getenv("AZURE_OPENAI_MODEL_NAME"),
17 "AZURE_OPENAI_API_KEY": os.getenv("AZURE_OPENAI_API_KEY"),
18 "AZURE_OPENAI_API_VERSION": os.getenv("AZURE_OPENAI_API_VERSION")

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS
Downloading certifi-2024.8.30-py3-none-any.whl (167 kB)
Downloading colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Downloading h11-0.14.0-py3-none-any.whl (58 kB)
Installing collected packages: typing-extensions, sniffio, jiter, idna, h11, distro, co
ydantic-core, httpcore, anyio, pydantic, httpx, openai
Successfully installed annotated-types-0.7.0 anyio-4.6.0 certifi-2024.8.30 colorama-0.4
httpx-0.27.2 idna-3.10 jiter-0.6.1 openai-1.51.2 pydantic-2.9.2 pydantic-core-2.23.4 s
s-4.12.2
PS C:\Users\BN743PF\Downloads\EYcode\EYcode> python -m venv venv
PS C:\Users\BN743PF\Downloads\EYcode\EYcode> venv\Scripts\activate
(venv) PS C:\Users\BN743PF\Downloads\EYcode\EYcode> |
```

Run the following command:

pip install -r requirements.txt

You may see error during the installation mentioning something like below:

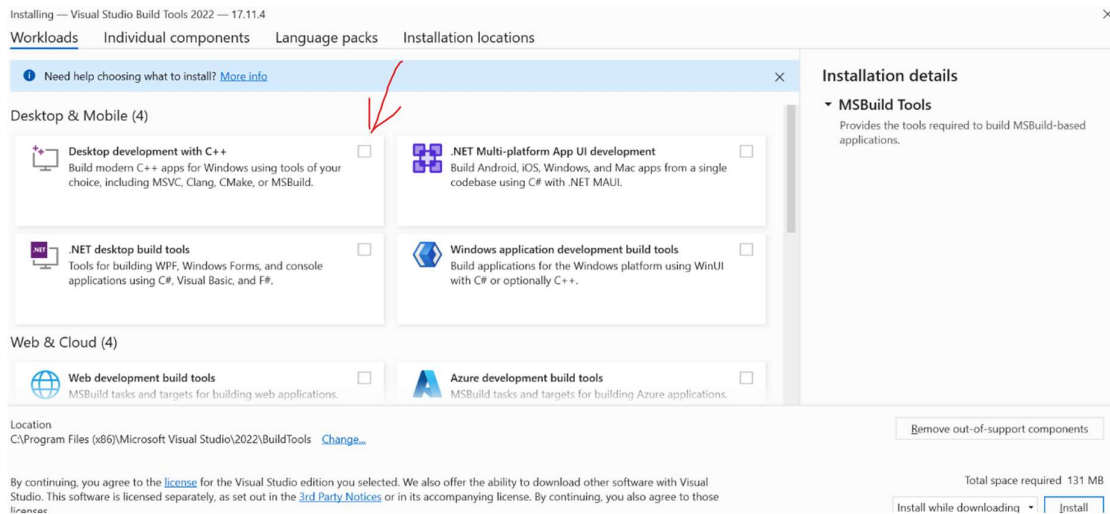
```
running bdist_wheel
running build
running build_ext
building 'hnsplib' extension
error: Microsoft Visual C++ 14.0 or greater is required. Get it with "Microsoft C++ Build Tools": https://visualstudio.mic
rosoft.com/visual-cpp-build-tools/
[end of output]
note: This error originates from a subprocess and is likely not a problem with pip
```

To fix it, install MS dev tools:

<https://visualstudio.microsoft.com/visual-cpp-build-tools/>

Name	Date modified	Type
Today (5)		
vs_BuildTools.exe	10-10-2024 02:32 PM	Application
tesla_db.zip	10-10-2024 12:24 PM	Compressed (zipped)
tesla-annual-reports.zip	10-10-2024 12:09 PM	Compressed (zipped)
index_tesla_documents.ipynb	10-10-2024 12:00 PM	Jupyter Source File
retrieval_tesla_documents.ipynb	10-10-2024 11:53 AM	Jupyter Source File
Yesterday (6)		

When you open the installer, select the first box below:



After installation, restart VS Code

And open the code folder.

Reactivate the virtual environment in the terminal:

In terminal, execute the following:

➤ `venv\Scripts\activate`

Install packages again:

`pip install -r requirements.txt`

Troubleshooting

- If terminal says scripts are disabled (PowerShell):
`Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser`
- If Python is not recognized, verify it's added to PATH
- Make sure to always activate virtual environment before installing packages. The active environment is shown in green as below:

Give command:

`venv\Scripts\activate`

```
45 # Create a twin axes for population and literacy rate
46 fig, ax1 = plt.subplots(figsize=(15, 8))

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE

1 Delhi 29000000 86.2
2 Bangalore 13000000 88.7
3 Hyderabad 9700000 83.3
4 Ahmedabad 8000000 89.1
PS D:\code\EYcode>
* History restored
PS D:\code\EYcode>
* History restored
PS D:\code\EYcode> venv\Scripts\activate
(venv) PS D:\code\EYcode>
```

- If still facing errors, try following and then run `pip install -r requirements.txt`:
`pip install --upgrade`
`pip setuptools wheel`
`pip cache purge`
`pip install --upgrade --force-reinstall setuptools`

Ater successful installation of packages you should see:

```
13 azure_config = {
14     "AZURE_OPENAI_ENDPOINT": os.getenv("AZURE_OPENAI_ENDPOINT"),
15     "AZURE_OPENAI_MODEL_DEPLOYMENT_NAME": os.getenv("AZURE_OPENAI_MODEL_DEPLOYMENT_NAME"),
16     "AZURE_OPENAI_MODEL_NAME": os.getenv("AZURE_OPENAI_MODEL_NAME"),
17     "AZURE_OPENAI_API_KEY": os.getenv("AZURE_OPENAI_API_KEY"),
18     "AZURE_OPENAI_API_VERSION": os.getenv("AZURE_OPENAI_API_VERSION")
}

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS Python +

Downloading certifi-2024.8.30-py3-none-any.whl (167 kB)
Downloading colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Downloading h11-0.14.0-py3-none-any.whl (58 kB)
Installing collected packages: typing-extensions, sniffio, jiter, idna, h11, distro, colorama, certifi, annot
ydantic-core, httpcore, anyio, pydantic, httpx, openai
Successfully installed annotated-types-0.7.0 anyio-4.6.0 certifi-2024.8.30 colorama-0.4.6 distro-1.9.0 h11-0.
httpx-0.27.2 idna-3.10 jiter-0.6.1 openai-1.51.2 pydantic-2.9.2 pydantic-core-2.23.4 sniffio-1.3.1 tqdm-4.66
s-4.12.2
PS C:\Users\BN743PF\Downloads\EYcode\EYcode> python -m venv venv
PS C:\Users\BN743PF\Downloads\EYcode\EYcode> venv\Scripts\activate
(venv) PS C:\Users\BN743PF\Downloads\EYcode\EYcode>
```

Thatsit!!

You are now ready to build your GenAI Python apps with VS code

Python essentials for this program

Download the python_basicapps.zip from:

https://github.com/vijay-agrawal/genaitraining/blob/main/code/00python_basicapps.zip

Unzip it into the **genaitraining** or your project root folder

In VS Code, you should see the folder and code, like below:

Using functions in python

The demo functions_demo.py can be reviewed to understand how function calling, parameter passing is done in Pythonn

Working with data tables using pandas

<https://www.youtube.com/watch?v=tRKeLrwfUgU>

Review the pandas demo code present in pandas_demo.py.

Run it by pressing the play button on top right.

Edit it to play with various dataframe capabilities such as sorting, adding a new derived column etc. Use help of your favourite AI chatbot to generate the code for the same and test it out.

Learn to use .env for API keys and private configuration

Best practice in python based application development is to keep private configuration parameters such as API Keys, passwords etc in a **.env** file

When your python application loads it can read the private key value pairs from the .env file and use them.

This gives flexibility and security and is a common practice to deal with private configuration parameters.

See the video below on how to setup a .env file for your project:

https://www.youtube.com/watch?v=8dlQ_nDE7dQ

Open the dotenv_demo.py in VS Code

Run it by using the “Play” button on top right in your VS Code.

Alternatively, you can open the terminal, activate your virtual environment

➤ Venv\scripts\activate

And issue following command:

➤ python dotenv_demo.py

The script should run and print the environment variables present in your .env file

Working with data files

Data files such as csv files can be easily loaded in Python. They are usually converted to pandas dataframe for further analysis.

Review the file reading demo code present in `dataloading.py`.

Run it by pressing the play button on top right.

You can check the data file it is loading, present in the data folder.

Using matplotlib for data visualization and analysis

Exploratory Data Analytics (EDA) with python

Run the `data_visualization.py` to see how matplotlib library helps with data visualization and analysis

A good video on how to do EDA with python:

<https://www.youtube.com/watch?v=xi0vhXFPegw>

Also search on Kaggle for existing notebooks people have made on the same.

<https://www.kaggle.com/code>

Accessing and consuming services using APIs

Review and run the code `weather_api_json.py`

It uses Weather API service from openweathermap (You can register to get your own free key)

Above should give a good handle on Python essentials for this program

Gen AI Basic app

You will now test a very basic Gen AI Python app, that will use Azure Open AI Keys to access an Open AI LLM hosted on Azure cloud.

Unzip the following into your project root folder:

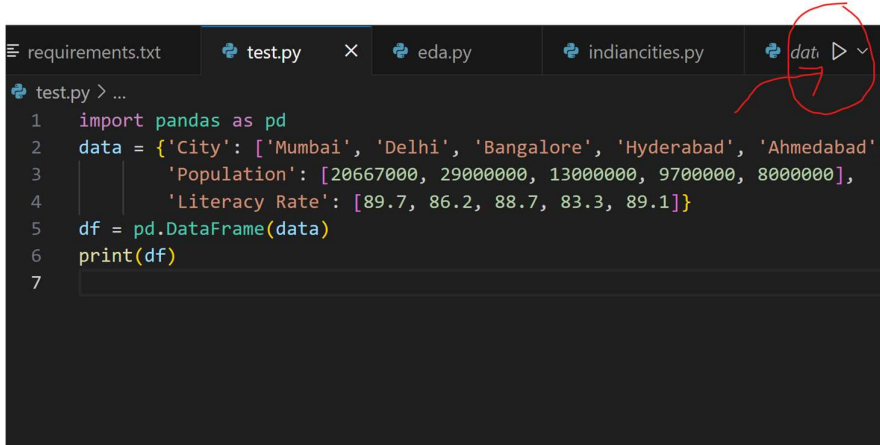
https://github.com/vijay-agrawal/genaitraining/blob/main/code/01azure_openai.zip

Unzip its contents into the genaitraining (or code) folder

Do not unzip into venv (Note that the virtual environment folder (.venv or venv) should not be touched once created)

Make sure you have `.env` file in your project root folder and it contains the Azure Open AI access keys.

Run the **test_azure_openai.py** by selecting it in the left pane and clicking on the **Play button** or by giving the command in terminal



```
test.py > ...
1 import pandas as pd
2 data = {'City': ['Mumbai', 'Delhi', 'Bangalore', 'Hyderabad', 'Ahmedabad'],
3         'Population': [20667000, 29000000, 13000000, 9700000, 8000000],
4         'Literacy Rate': [89.7, 86.2, 88.7, 83.3, 89.1]}
5 df = pd.DataFrame(data)
6 print(df)
7
```

If unable to run, you may need to select Python Interpreter:

- Press Ctrl+Shift+P (Cmd+Shift+P on macOS)
- Type "Python: Select Interpreter"
- Choose the interpreter from your virtual environment

Also ensure your virtual environment is enabled in the terminal.

In VS Code, open terminal and type the following if not enabled

➤ `venv\scripts\activate`

You should see the output properly showing the response from the LLM

Try to give different prompts such as “Who is the Prime Minister of India” or “Who is the Chief Minister of Maharashtra” or “Tell me about latest results of TCS” etc.

You are now ready to explore the exciting world of AI Apps in the classroom!